

Turning an unencrypted, unlogged AWS environment into a CIS-benchmarked cloud estate

A leading financial institution engaged Sthiros for a comprehensive security assessment of its AWS infrastructure, covering 19 control areas across identity, network, and application security.

Client
A Leading Financial Institution

Product Deployed
AWS Cloud Security
Assessment & Controls Review

Sector
Banking & Financial Services

Geography
Global

The Challenge:

When a bank's AWS environment has never been benchmarked

This financial institution needed a comprehensive security assessment of its AWS infrastructure to identify misconfigurations, validate security controls, and confirm compliance with banking regulations and cloud security best practices.



Encryption & Data Protection

VPN traffic, S3 buckets, API calls, and inbound/outbound traffic all ran without encryption, leaving data exposed at rest and in transit.



Logging & Monitoring

CloudWatch and CloudTrail alerts were never set, VPC flow logs were disabled, and CloudFront and ELB logging was inactive, leaving anomalies undetectable.



Access Control & Authentication

API calls could proceed without authorisation, console downloads created an exfiltration path, and key rotation was never enabled.



Security Controls & Architecture

No VAPT of the AWS infrastructure had ever been performed, with no hardening standard followed and no network architecture diagram available.

The Solution:

A structured review across every layer of the AWS estate

Sthiros reviewed 19 AWS control areas across identity, network, application, and DevOps layers, benchmarking every finding against the CIS AWS Foundations Benchmark and financial services cloud security standards.

Every one of the 16 gaps identified was paired with a specific remediation, from encryption enforcement to a quarterly VAPT schedule, not a generic hardening checklist.

The Methodology



Identity & Access Control Review

AWS Root Account, IAM, access procedures, and password policy assessed for MFA usage, resource-level permissions, and credential hygiene.



Network & Infrastructure Security Review

Security Groups, NACLs, routing tables, VPC isolation, and QA/Production segregation assessed for encryption, monitoring, and access control.



Application & Edge Security Review

AWS WAF, ELB/ECR, CloudFront, and Autoscaling reviewed for HTTPS enforcement, DDoS protection, and IP-level controls.



DevOps & Change Control Review

Jenkins authentication and authorisation, change management approvals, and pre-production checklists assessed for governance gaps.



Remediation & Hardening

Every finding remediated against the CIS AWS Foundations Benchmark, with quarterly VAPT scheduled going forward.

What This Means

Encrypt data everywhere, always.

See threats as they happen.

Audit the cloud like a bank.

Outcomes

19

AWS control areas reviewed

Spanning IAM, encryption, logging, network isolation, and application security.

16

Security gaps identified

Across encryption, logging, access control, and architecture documentation.

13

Remediation actions completed

From VPN encryption to a quarterly VAPT schedule going forward.

90-Day

Key rotation cycle enforced

Replacing long-lived credentials with a defined rotation policy.

Get in touch

Is your cloud environment benchmarked, or just deployed?

Speak to Sthiros about what a structured AWS security review can unlock for your cloud.

Connect with us:
sthiros.com

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